MOHAMMED NAFEES H

Robotics Engineering / Robotics Software / Embedded Robotics Intern

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PROFESSIONAL SUMMARY

B.Tech Robotics & Artificial Intelligence student at SASTRA University, Class of 2028, targeting robotics engineering, robotics software, and embedded robotics internships. Hands-on experience across ROS 2 workflows, Python/C++ robotics programming, embedded hardware, sensors, motor control, computer vision, and AI backend pipelines. Engineered a 4x4 Arduino-Python autonomous navigation robot, a 3-class fruit quality grading system, and a 3-stage YOLO/OpenCV/BLIP image-analysis backend for a larger Visage project.

EDUCATION

SASTRA Deemed University | B.Tech, Robotics & Artificial Intelligence | Aug 2024 - May 2028 | Tamil Nadu, India

TECHNICAL SKILLS

- Robotics Software: ROS 2, Python, C/C++, Linux/Ubuntu, Git, robot integration, A* path planning, controls basics, sensor debugging
- Embedded Hardware: Arduino, ESP32, STM32, Raspberry Pi 5, Raspberry Pi Zero 2 W, Pi Camera, GPIO, motor drivers, encoders, IR sensors
- Communication & Sensors: I2C, UART, LiDAR, IMU, camera sensors, serial communication, sensor/actuator interfacing
- Hardware Lab Exposure: soldering, schematic reading, hardware bring-up, wiring, component evaluation, procurement coordination
- AI & Computer Vision: OpenCV, YOLOv8, Ultralytics, TensorFlow/Keras, CLIP, BLIP, Hugging Face Transformers, image classification, object detection, scene captioning
- Tools: VS Code, Ubuntu 24.04, CoppeliaSim, Fusion 360, EasyEDA/KiCad basics

EXPERIENCE

Industrial Robotics Engineering Project | Robotics Engineering Contributor | ROS 2, Controls, Embedded Integration

- Contribute to ROS 2 based robotics workflows, controls-oriented development, embedded integration, sensor/actuator evaluation, and hardware coordination under professional constraints.
- Coordinate component identification, evaluation, and procurement support for a project hardware budget of approximately Rs. 10 lakh.
- Work across robotics software, electronics, and mechanical constraints for deployment-oriented industrial robotics tasks.

PROJECTS

Hadi Al Hajatron | Arduino, Python, IR Sensors, Encoders, A* Path Planning

GitHub: https://github.com/mohammednafees1007-hub/Hadi-Al-Hajatron

- Engineered a 4x4 Arduino-Python autonomous grid-navigation robot with 3 IR obstacle sensors, 60 cm calibrated movement steps, live mapping, map export, and A* route planning.

FruitVision AI - Fruit Quality Detector for SMS | YOLOv8x, Keras, CLIP, OpenCV

GitHub: https://github.com/mohammednafees1007-hub/fruit-quality-detector-for-sms

- Developed a quality-first fruit grading system with image input workflows, YOLOv8x fruit-region detection, a 3-class Keras quality model, A/B/C grade mapping, and annotated result output.

AI Image Analyzer | Python, YOLOv8n, OpenCV, BLIP, Transformers

GitHub: https://github.com/mohammednafees1007-hub/ai-image-analyzer

- Created a 3-stage backend image-analysis pipeline for a larger Visage project with YOLOv8n object detection, 13 allowed object classes, privacy-safe face tokens, BLIP captions, and structured app-ready output.

Quadruped Robot | Inverse Kinematics, Robotics Controls, Embedded Integration

- Implemented inverse-kinematics motion logic for leg movement, connecting control theory with practical actuation for stable quadruped locomotion.

Schooling-Era Robotics Projects | Arduino, Sensors, Motor Drivers

- Prototyped early Arduino robotics systems involving sensors, motor drivers, wiring, control experiments, and debugging.

LEADERSHIP AND INVOLVEMENT

Robotics Club Member | SASTRA Deemed University

- Active in robotics learning, project building, experimentation, and applied technical development.

ENGINEERING HIGHLIGHTS

- Practical builder across robotics software, embedded electronics, mechanical integration, AI models, and user-facing tools.
- Comfortable with Linux robotics environments, hardware bring-up, sensor debugging, serial communication, rapid prototyping, and deployment-oriented project work.
- Strong project evidence through public GitHub repositories, runnable local applications, backend pipelines, and real hardware workflows.